

Ammar Yasser Abdallah

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AI & Machine Learning Engineer | Intelligent Systems Developer

Professional Summary

Artificial Intelligence graduate and hands-on developer focused on applied ML, NLP/RAG, computer vision, wearable-signal analysis, robotics, and AI-enabled applications. Experienced with Python, scikit-learn, Pandas, NumPy, OpenCV, MediaPipe, YOLOv8 concepts, LangGraph, LangChain, Gemini API, ChromaDB, BM25 retrieval, Java/JavaFX AI search, Arduino/C++, ESP32, and full-stack integration using React, Next.js, NestJS, MongoDB, and Socket.IO.

Education

Bachelor's Degree - Computers and Data Science, Artificial Intelligence Department
Alexandria University, Faculty of Computers and Data Science | Intelligent Systems Department | Graduated 2026

AI & Technical Skills

- ML/Data: Python, scikit-learn, Pandas, NumPy
- Signal AI: HRV, PPG/BVP, feature extraction
- NLP/RAG: LangGraph, LangChain, Gemini API
- Retrieval: ChromaDB, BM25, Reciprocal Rank Fusion
- Computer Vision: OpenCV, MediaPipe, YOLOv8
- Algorithms: A*, UCS, Minimax, Alpha-Beta
- Models: Decision Trees, Random Forest, XGBoost
- Metrics: accuracy, F1, AUC, Cohen's kappa, MAE
- Robotics/IoT: Arduino/C++, ESP32, Raspberry Pi
- Backend AI APIs: FastAPI concepts, NestJS integration
- Data systems: MongoDB, SQL, Mongoose, APIs
- Tools: GitHub, Docker, Postman, MATLAB, Figma

Graduation Project - Detailed AI Profile

Mind Guard - AI-Powered Mental Health Monitoring Ecosystem

Project: Team graduation project, Alexandria University, 2025-2026 | Wearable IoT, behavioral sensing, ML, and RAG mental-health support

AI Stack: Python, scikit-learn, Pandas, NumPy, NeuroKit2 concepts, LangGraph, LangChain, Gemini, Vertex AI, ChromaDB, BM25, TF-IDF, PyTorch, FastAPI-style model services

- Contributed to an AI-powered mental health ecosystem designed to move care from reactive checkups to continuous monitoring using wearable HRV/PPG signals, mobile behavior patterns, self-assessment scales, AI risk insights, and doctor intervention workflows.
- Worked with standardized mental-health assessments and screening signals including PHQ-9, GAD-7, PSS/DASS-style stress framing, HRV features, sleep signals, and behavioral indicators to support explainable patient monitoring.
- Validated the HRV stress-classification path using the SWELL-KW dataset, mapping 30-second physiological windows into no-stress, moderate-stress/time-pressure, and severe-stress/interruption classes.
- Documented a deterministic signal-processing pipeline for raw PPG windows: high-pass filtering, z-score normalization, heartbeat peak detection, RR-interval construction, and HRV feature extraction across time-domain, frequency-domain, and non-linear metrics.
- Reported stress-model performance around 0.90 hold-out accuracy, 0.899 5-fold cross-validation accuracy, and Cohen's kappa around 0.83, supporting strong agreement beyond chance while keeping claims framed as wellness screening rather than diagnosis.
- Used SWELL-KW as the primary cognitive-load dataset and WESAD as a wearable benchmark for wrist-worn BVP/PPG, ECG, EDA, temperature, and affective-state validation.
- Helped define a multi-model intelligence layer with an HRV stress classifier, sleep-stage classifier, GAD-7 anxiety screening model, and transparent rule-based correlation engine producing risk level, flags, predictions, and recommendations.
- Designed NEURA as a safety-aware RAG chatbot: a standalone Python service with a ten-node LangGraph pipeline covering guard, language detection, crisis checks, history compression, routing, query rewriting, retrieval, generation, and post-processing.
- Implemented hybrid retrieval concepts using ChromaDB dense vector search, BM25 sparse retrieval, Reciprocal Rank Fusion, section re-ranking, and top-3 context assembly to ground responses in curated mental-health knowledge.
- Added safety-oriented conversational behavior including English/Arabic crisis keyword scanning, Gemini soft-signal crisis confirmation, non-diagnostic prompt boundaries, response post-processing, feedback scoring, and session-title generation.

Additional AI, Robotics & Computer Vision Projects

VisionDeck Computer Vision Suite

Python, Streamlit, OpenCV, MediaPipe, YOLOv8, Webcam/IP Camera

- Developed a computer-vision dashboard combining face detection, hand gesture recognition, real-time object detection, webcam/IP camera support, and YOLOv8 coursework fine-tuning workflows.

Biped Robot Control System

Python, Arduino/C++, ESP32, Streamlit, computer vision, robotics

- Worked on a six-servo ESP32 biped robot with desktop control, Streamlit diagnostics, pose editing, joint calibration, color-based robot/ball tracking, and autonomous forward movement until collision.

DUM-E Robotic Arm System

Python, Arduino/C++, ESP32, Streamlit, FK/IK, ROS2-style dashboard

- Built and used a robot-arm control stack with firmware, endpoint discovery, forward/inverse kinematics helpers, sequence replay, and guarded pick-and-place execution.

KBAI Book Chatbot

Python, Google Gemini API, Retrieval-Augmented Generation, NLP

- Built a knowledge-based AI chatbot for course content using Gemini API and RAG concepts to retrieve relevant context and generate explainable answers.

Search, Optimization & Intelligent Games

Java, JavaFX, Minimax, Alpha-Beta, A*, UCS, Greedy Best-First Search, ACO/GA concepts

- Built Connect 4 AI, 8-Puzzle Solver, and TSP-style optimization projects using adversarial search, heuristic search, path optimization, and custom evaluation strategies.

Machine Translation: English-Spanish

Python, NLP, RNN/Transformer concepts, tokenization, BLEU evaluation

- Built an NLP translation project with cleaned parallel text pipelines, sequence-to-sequence concepts, transformer concepts, and translation evaluation.

Full-Stack Engineering Foundation

React, Next.js, TypeScript, Node.js, Express.js, NestJS, MongoDB, Mongoose, Socket.IO, REST APIs, JWT authentication, Tailwind CSS, responsive UI, EventX Studio, Saraha backend API, Movies API website, and frontend portfolio projects.

Courses

Courses: Machine Learning; Natural Language Processing; Pattern Recognition; Databases; Cloud Computing; Operations Research; Embedded Systems; Internet of Things; Modern Control Systems; Full-Stack Web Development - Route Academy.

Languages

Arabic - Native | English - Good working proficiency